Create a database named Assignment_Database.

```
mysql> CREATE DATABASE Assignment_Database;
Query OK, 1 row affected (0.05 sec)
```

Create a schema named assignment_1.

```
mysql> CREATE SCHEMA assignment_1;
Query OK, 1 row affected (0.00 sec)
```

employees table:

CREATE TABLE employees (

employee_id INT PRIMARY KEY,

first_name VARCHAR(50),

last_name VARCHAR(50),

hire_date DATE,

department_id INT,

salary DECIMAL(10, 2)

);

```
mysql> DESC assignment_1.employees;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| employee_id   | int(11)       | NO   | PRI | NULL    |       |
| first_name    | varchar(50)   | YES  |     | NULL    |       |
| last_name     | varchar(50)   | YES  |     | NULL    |       |
| hire_date     | date          | YES  |     | NULL    |       |
| department_id | int(11)       | YES  |     | NULL    |       |
| salary        | decimal(10,2) | YES  |     | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
```

departments table:

CREATE TABLE departments (

department_id INT PRIMARY KEY,

department_name VARCHAR(100)

);

```
mysql> create table assignment_1.departments (
    ->
    -> department_id INT PRIMARY KEY,
    ->
    -> department_name VARCHAR(100)
    ->
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> Desc assignment_1.departments;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| department_id   | int(11)      | NO   | PRI | NULL    |       |
| department_name | varchar(100) | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
2 rows in set (0.02 sec)
```

INSERT INTO departments (department_id, department_name)

VALUES

(1, 'Sales'),

(2, 'Marketing'),

(3, 'Engineering'),

(4, 'HR'),

(5, 'Finance');

```
mysql> SELECT * FROM assignment_1.departments;
+---------------+-----------------+
| department_id | department_name |
+---------------+-----------------+
|             1 | Sales           |
|             2 | Marketing       |
|             3 | Engineering     |
|             4 | HR              |
|             5 | Finance         |
+---------------+-----------------+
5 rows in set (0.03 sec)
```

INSERT INTO employees (employee_id, first_name, last_name, hire_date, department_id, salary) VALUES

(1, 'John', 'Doe', '2015-06-23', 1, 55000.00),

(2, 'Jane', 'Smith', '2018-02-10', 2, 62000.00),

(3, 'Samuel', 'Adams', '2012-11-04', 3, 90000.00),

(4, 'Emily', 'Clark', '2020-03-15', 1, 45000.00),

(5, 'Daniel', 'Harris', '2016-07-19', 4, 49000.00),

(6, 'Rachel', 'Baker', '2019-10-01', 3, 95000.00),

(7, 'Paul', 'Jones', '2017-09-13', 2, 55000.00),

(8, 'Sophia', 'Taylor', '2014-12-21', 5, 73000.00),

(9, 'Michael', 'Lee', '2011-08-14', 4, 47000.00),

(10, 'Olivia', 'King', '2022-01-30', 3, 98000.00);

```
mysql> INSERT INTO assignment_1.employees (employee_id, first_name, last_name, hire_date, department_id, salary) VALUES
    -> (1, 'John', 'Doe', '2015-06-23', 1, 55000.00),
    -> (2, 'Jane', 'Smith', '2018-02-10', 2, 62000.00),
    -> (3, 'Samuel', 'Adams', '2012-11-04', 3, 90000.00),
    -> (4, 'Emily', 'Clark', '2020-03-15', 1, 45000.00),
    -> (5, 'Daniel', 'Harris', '2016-07-19', 4, 49000.00),
    -> (6, 'Rachel', 'Baker', '2019-10-01', 3, 95000.00),
    -> (7, 'Paul', 'Jones', '2017-09-13', 2, 55000.00),
    -> (8, 'Sophia', 'Taylor', '2014-12-21', 5, 73000.00),
    -> (9, 'Michael', 'Lee', '2011-08-14', 4, 47000.00),
    -> (10, 'Olivia', 'King', '2022-01-30', 3, 98000.00);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> select * from assignment_1.employees;
+-------------+------------+-----------+------------+---------------+----------+
| employee_id | first_name | last_name | hire_date  | department_id | salary   |
+-------------+------------+-----------+------------+---------------+----------+
|           1 | John       | Doe       | 2015-06-23 |             1 | 55000.00 |
|           2 | Jane       | Smith     | 2018-02-10 |             2 | 62000.00 |
|           3 | Samuel     | Adams     | 2012-11-04 |             3 | 90000.00 |
|           4 | Emily      | Clark     | 2020-03-15 |             1 | 45000.00 |
|           5 | Daniel     | Harris    | 2016-07-19 |             4 | 49000.00 |
|           6 | Rachel     | Baker     | 2019-10-01 |             3 | 95000.00 |
|           7 | Paul       | Jones     | 2017-09-13 |             2 | 55000.00 |
|           8 | Sophia     | Taylor    | 2014-12-21 |             5 | 73000.00 |
|           9 | Michael    | Lee       | 2011-08-14 |             4 | 47000.00 |
|          10 | Olivia     | King      | 2022-01-30 |             3 | 98000.00 |
+-------------+------------+-----------+------------+---------------+----------+
```

1) To concat the first_name and last_name of employees into a single column named full_name.

```
mysql> SELECT CONCAT(first_name,' ',last_name) AS full_name from assignment_
1.employees;
+---------------+
| full_name     |
+---------------+
| John Doe      |
| Jane Smith    |
| Samuel Adams  |
| Emily Clark   |
| Daniel Harris |
| Rachel Baker  |
| Paul Jones    |
| Sophia Taylor |
| Michael Lee   |
| Olivia King   |
+---------------+
10 rows in set (0.00 sec)
```

3) Query counts the number of employees in each department.

```
mysql> SELECT department_id, COUNT(*) AS employee_count
    -> FROM assignment_1.employees
    -> GROUP BY department_id;
+---------------+----------------+
| department_id | employee_count |
+---------------+----------------+
|             1 |              2 |
|             2 |              2 |
|             3 |              3 |
|             4 |              2 |
|             5 |              1 |
+---------------+----------------+
5 rows in set (0.02 sec)
```

4) Query to lists all departments and counts the number of employees in each department

```
mysql> SELECT d.department_name, COUNT(e.employee_id) AS employee_count
    -> FROM departments d
    -> LEFT JOIN employees e
    -> ON d.department_id = e.department_id
    -> GROUP BY d.department_name;
+----------------+---------------+
| department_name | employee_count |
+----------------+---------------+
| Engineering    |             3 |
| Finance        |             1 |
| HR             |             2 |
| Marketing      |             2 |
| Sales          |             2 |
+----------------+---------------+
5 rows in set (0.00 sec)
```

5) To find employees with salary greater than average salary of their department

```
mysql> SELECT
    ->     e.first_name,
    ->     e.last_name,
    ->     e.salary,
    ->     e.department_id
    -> FROM employees e
    -> WHERE e.salary > (
    ->     SELECT AVG(e2.salary)
    ->     FROM employees e2
    ->     WHERE e2.department_id = e.department_id
    -> );
+------------+-----------+----------+---------------+
| first_name | last_name | salary   | department_id |
+------------+-----------+----------+---------------+
| John       | Doe       | 55000.00 |             1 |
| Jane       | Smith     | 62000.00 |             2 |
| Daniel     | Harris    | 49000.00 |             4 |
| Rachel     | Baker     | 95000.00 |             3 |
| Olivia     | King      | 98000.00 |             3 |
+------------+-----------+----------+---------------+
5 rows in set (0.01 sec)
```

6) To get all employee names in each department.

```
mysql> SELECT
    ->     d.department_name,
    ->     GROUP_CONCAT(CONCAT(e.first_name, ' ', e.last_name)) AS employee_
names
    -> FROM departments d
    -> LEFT JOIN employees e
    -> ON d.department_id = e.department_id
    -> GROUP BY d.department_name;
+----------------+-------------------------------------+
| department_name | employee_names                      |
+----------------+-------------------------------------+
| Engineering     | Rachel Baker,Samuel Adams,Olivia King |
| Finance         | Sophia Taylor                       |
| HR              | Michael Lee,Daniel Harris           |
| Marketing       | Paul Jones,Jane Smith               |
| Sales           | John Doe,Emily Clark                |
+----------------+-------------------------------------+
5 rows in set (0.00 sec)
```

8) Group employees by hire year and calculate the total salary for each year

```
mysql> SELECT
    ->     YEAR(hire_date) AS hire_year,
    ->     SUM(salary) AS total_salary
    -> FROM employees
    -> GROUP BY YEAR(hire_date);
+-----------+--------------+
| hire_year | total_salary |
+-----------+--------------+
|      2011 |     47000.00 |
|      2012 |     90000.00 |
|      2014 |     73000.00 |
|      2015 |     55000.00 |
|      2016 |     49000.00 |
|      2017 |     55000.00 |
|      2018 |     62000.00 |
|      2019 |     95000.00 |
|      2020 |     45000.00 |
|      2022 |     98000.00 |
+-----------+--------------+
10 rows in set (0.00 sec)
```

9) Group departments and count the number of employees in each department

```
mysql> SELECT
    ->     d.department_name,
    ->     COUNT(e.employee_id) AS employee_count
    -> FROM departments d
    -> LEFT JOIN employees e
    -> ON d.department_id = e.department_id
    -> GROUP BY d.department_name;
+-----------------+----------------+
| department_name | employee_count |
+-----------------+----------------+
| Engineering     |              3 |
| Finance         |              1 |
| HR              |              2 |
| Marketing       |              2 |
| Sales           |              2 |
+-----------------+----------------+
5 rows in set (0.00 sec)
```

10) Find the highest salary in each department

```
mysql> SELECT
    ->     department_id,
    ->     MAX(salary) AS highest_salary
    -> FROM employees
    -> GROUP BY department_id;
+---------------+----------------+
| department_id | highest_salary |
+---------------+----------------+
|             1 |       55000.00 |
|             2 |       62000.00 |
|             3 |       98000.00 |
|             4 |       49000.00 |
|             5 |       73000.00 |
+---------------+----------------+
5 rows in set (0.00 sec)
```

11) Group employees by department and show the average salary in each department.

```
mysql> SELECT
    ->     department_id,
    ->     AVG(salary) AS avg_salary
    -> FROM employees
    -> GROUP BY department_id;
+---------------+--------------+
| department_id | avg_salary   |
+---------------+--------------+
|             1 | 50000.000000 |
|             2 | 58500.000000 |
|             3 | 94333.333333 |
|             4 | 48000.000000 |
|             5 | 73000.000000 |
+---------------+--------------+
5 rows in set (0.01 sec)
```

12) Find departments with more than 5 employees

```
mysql> SELECT
    ->     d.department_name,
    ->     COUNT(e.employee_id) AS employee_count
    -> FROM departments d
    -> LEFT JOIN employees e
    -> ON d.department_id = e.department_id
    -> GROUP BY d.department_name;
+----------------+---------------+
| department_name | employee_count |
+----------------+---------------+
| Engineering     |             3 |
| Finance         |             1 |
| HR              |             2 |
| Marketing       |             2 |
| Sales           |             2 |
+----------------+---------------+
5 rows in set (0.00 sec)
```

13) List departments where the average salary is greater than 50,000.

```
mysql> SELECT
    ->     d.department_name,
    ->     AVG(e.salary) AS avg_salary
    -> FROM employees e
    -> JOIN departments d
    -> ON e.department_id = d.department_id
    -> GROUP BY d.department_name
    -> HAVING AVG(e.salary) > 50000;
+----------------+--------------+
| department_name | avg_salary   |
+----------------+--------------+
| Engineering     | 94333.333333 |
| Finance         | 73000.000000 |
| Marketing       | 58500.000000 |
+----------------+--------------+
3 rows in set (0.00 sec)
```

14) List employees who earn more than 60,000 and belong to a department with more than 3 employees.

```
mysql> SELECT
    ->     e.first_name,
    ->     e.last_name,
    ->     e.salary,
    ->     e.department_id
    -> FROM employees e
    -> WHERE e.salary > 60000
    -> AND e.department_id IN (
    ->     SELECT department_id
    ->     FROM employees
    ->     GROUP BY department_id
    ->     HAVING COUNT(employee_id) > 3
    -> );
Empty set (0.00 sec)
```

15) Show departments where there is more than one employee with a salary over 100,000.

```
mysql> SELECT
    ->     d.department_name,
    ->     COUNT(e.employee_id) AS high_salary_count
    -> FROM employees e
    -> JOIN departments d
    -> ON e.department_id = d.department_id
    -> WHERE e.salary > 100000
    -> GROUP BY d.department_name
    -> HAVING COUNT(e.employee_id) > 1;
Empty set (0.00 sec)
```

16) Delete an employee with a specific employee_id (e.g., 5)

```
mysql> DELETE FROM employees
    -> WHERE employee_id = 5;
Query OK, 1 row affected (0.00 sec)

mysql> select * from employees
    -> ;
+-------------+------------+-----------+------------+---------------+----------+
| employee_id | first_name | last_name | hire_date  | department_id | salary   |
+-------------+------------+-----------+------------+---------------+----------+
|           1 | John       | Doe       | 2015-06-23 |             1 | 55000.00 |
|           2 | Jane       | Smith     | 2018-02-10 |             2 | 62000.00 |
|           3 | Samuel     | Adams     | 2012-11-04 |             3 | 90000.00 |
|           4 | Emily      | Clark     | 2020-03-15 |             1 | 45000.00 |
|           6 | Rachel     | Baker     | 2019-10-01 |             3 | 95000.00 |
|           7 | Paul       | Jones     | 2017-09-13 |             2 | 55000.00 |
|           8 | Sophia     | Taylor    | 2014-12-21 |             5 | 73000.00 |
|           9 | Michael    | Lee       | 2011-08-14 |             4 | 47000.00 |
|          10 | Olivia     | King      | 2022-01-30 |             3 | 98000.00 |
+-------------+------------+-----------+------------+---------------+----------+
9 rows in set (0.00 sec)
```

17) Delete employees who have a salary lower than 30,000

```
mysql> DELETE FROM employees
    -> WHERE salary < 30000;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from employees;
+-------------+------------+-----------+------------+---------------+----------+
| employee_id | first_name | last_name | hire_date  | department_id | salary   |
+-------------+------------+-----------+------------+---------------+----------+
|           1 | John       | Doe       | 2015-06-23 |             1 | 55000.00 |
|           2 | Jane       | Smith     | 2018-02-10 |             2 | 62000.00 |
|           3 | Samuel     | Adams     | 2012-11-04 |             3 | 90000.00 |
|           4 | Emily      | Clark     | 2020-03-15 |             1 | 45000.00 |
|           6 | Rachel     | Baker     | 2019-10-01 |             3 | 95000.00 |
|           7 | Paul       | Jones     | 2017-09-13 |             2 | 55000.00 |
|           8 | Sophia     | Taylor    | 2014-12-21 |             5 | 73000.00 |
|           9 | Michael    | Lee       | 2011-08-14 |             4 | 47000.00 |
|          10 | Olivia     | King      | 2022-01-30 |             3 | 98000.00 |
+-------------+------------+-----------+------------+---------------+----------+
9 rows in set (0.00 sec)
```